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NewJeans Stem Player

The project is an interactive NewJeans stem player that lets users browse the group's releases, select a song, and listen to individual audio stems such as vocals, instruments, drums, and bass. The goal is to create a playful, fan-focused listening experience where users can explore how each song is built.

Figma Mock up: <https://www.figma.com/design/1i4CLZvhwh8n1aZinX0hT5/NewJeans-Stem-Player--Mock-Up---Copy-?m=auto&t=wSWC1ePbYvEpwIqa-6>

Product Overview

The NewJeans Stem Player is designed as a playful music exploration tool. Users start by browsing NewJeans releases, selecting an EP, album, or single; choosing a song from that release; and then opening the main stem player workspace. The experience is built around three major layers: release browsing, song selection, and stem-based playback. The strongest part of the concept is that it combines fan identity with music analysis functionality. It not only plays a track, but it also lets users see who is singing, follow lyrics in multiple formats, and isolate parts of the arrangement.

Primary User Flow

Home Page -> Release Selection -> Song Selection Page -> Stem Player Page

Home Page



The home page displays all NewJeans releases in reverse chronological order, from newest to oldest. Users can scroll through the release list and select an EP, album, or single to open its song selection page.

Each release should appear as a clickable card containing the cover artwork, release title, release year, release type, and song count. The select/play button on each card should serve as the primary call to action.

The bottom-left icon and bottom-right link connect to the creator’s Twitter/X account and Linktree. These links should remain secondary to the main music-navigation controls.

Home Page UI Elements

Element	Purpose	Recommended Detail
Release cards	Allow the user to choose an EP, album, or single.	Include artwork, title, year, release type, and song count.
Scroll area	Lets the user browse the full release list.	Keep the newest releases at the top.
Card action button	Opens the selected release.	Simple Play Icon.
Creator links		Keep small and visually secondary.

Connect to Twitter/X and
Linktree.

Use case:

Users begin on the home page, where all NewJeans releases are listed from newest to oldest. Each release card includes the cover image, title, release year, release type, and number of songs. Selecting a release opens the song selection page for that specific EP, album, or single.

Song Selection Page



The song selection page lets users choose a song from the selected release. It uses a carousel-style interface: the pink left and right arrow buttons move between songs, while the main artwork and title update to show the currently selected song.

The centered "Select" button confirms the current song and opens the stem player page. The home icon in the top-left corner returns users to the main release page.

Song Selection UI Elements

Element	Purpose	Recommended Detail
Song artwork	Shows the current selected song or release artwork.	Include artwork, title, year, release type, and song count.

Pink arrows	Cycle between songs in the selected release.	Add hover labels: Previous Song and Next Song.
Select button	Opens the Stem Player.	Rename to "Open Stem Player" for stronger clarity.
Home icon	Returns to the home page.	Add hover label.

Use case:

This page uses a carousel-style selector. The pink arrows allow users to cycle through songs from the selected release, while the Select button opens the currently displayed song in the stem player.

Navigation Rule: The song selection page should only handle choosing songs from the selected release. Deeper playback controls, lyric display, stem mixing, and song metadata belong on the stem player page.

Stem Player Page



The stem player page is the main interactive workspace. It allows users to play the selected song, follow timed lyrics, identify which member is singing, and control individual audio stems. The top-left section displays the song title, song length, musical key, and BPM.

To the right, a timed lyrics box shows the current lyric line in sync with the music. The language selector allows users to switch between English, Korean, and romanized lyrics.

Beside the lyric line, an emoji indicates which member is currently singing. The member portraits on the left update in real time, with the active singer highlighted and inactive members grayed out.

Member Indicator System

Indicator	Member / Meaning
Teddy bear 🧸	Minji
Otter 🦦	Hanni
Dog 🐕	Danielle
Cat 🐈	Haerin
Hamster 🐹	Hyein
Bunny 🐰	Full group or unison vocal section

Mixer Workspace

The center of the page contains the Interactive Multi-Track Audio Mixer Workspace. Each row represents a separate audio stem, including vocals, instruments, drums, and bass. Every row includes a waveform display, synchronized playhead, mute and solo controls, and an individual volume slider.

At the bottom, the main audio timeline scrubber shows the song’s playback position and allows users to jump to different points in the track. The scrubber moves in sync with the playhead across all stem rows.

Song Navigation Menu

The icon in the top-right corner opens the song navigation menu. This menu lets users switch to another song from the same EP, album, or single, move to the previous or next song, or return to the song selection page.

Song Navigation Menu:

- Back to Song Selection
- Previous Song
- Next Song

Stem Playback Model

Default playback: Vocals + Instruments + Drums + Bass = Full mix

The player should use the separated stems as the main playback source. By default, all stems play together at the same time to recreate the full song mix. Each stem row controls its own audio file

Stem Row Controls

Control	Function	Visual Feedback
Volume slider	Adjusts the level of that individual stem.	Show value feedback, such as a percentage or level indicator.
Mute	Removes that stem from the mix	The button turns red
Solo	Isolates one or more selected stems.le	The button turns green
Playhead	Moves in sync across all stems.	
Waveform	Visualizes that specific stem	Seeking: Click anywhere on the waveform to jump to that position in the track. Time Preview: Hover over the waveform to see the time at that position.

Use case:

The stem player uses separated audio stems as the main playback source. By default, all stems play together in sync to recreate the full song mix. Each row controls one stem category, such as vocals, instruments, drums, or bass. Users can adjust the volume of each stem, mute individual stems, or solo one or more stems to isolate specific parts of the song. All stems share the same playback timeline and synchronized playhead, so changes happen in real time while the song plays. A master audio file may be used as a fallback only if stem playback is unavailable.

Technical and Implementation Notes

Core Systems

System	Role
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Web Audio API	Stem playback, gain control, mute, solo, and synchronized audio routing.
Wavesurfer.js	Waveform rendering and visual playback tracking.
Tone.js	Timing, transport logic, and scheduling if more precise sequencing is needed.
JSON metadata	Stores song info, lyric timestamps, language variants, and member indicators.
React or Next.js	Front-end application structure.
Tailwind CSS	Fast styling and responsive layout.
Zustand or Redux	Shared playback state, selected song, active stems, lyric state, and navigation state.